



Single Cell Perturbation Modeling with Geometry-Regularized Generative Flows

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Introduction

Machine learning in single cell transcriptomic data enables predictions of drug perturbation effects, reducing wet-lab cost and accelerating scientific discovery. One key challenge is generalizing to underrepresented contexts (new donors, cell types with limited data).

Generative flow models, like Flow Matching (FM) and Neural ODEs allow for generation of distributions perturbed cells and geometry-informed machine learning has enabled improvements in modeling of cell trajectories. We explore if combining these two ideas, generative flow models for perturbation prediction and geometric-regularized latent spaces, can improve flow-model performance by aligning flows with paths on the manifold of the data.

Key Contributions

- > Use **Flow Matching** and **Neural ODE** models in autoencoder latent space for single-cell perturbation prediction
- > Show **geometric regularization** of autoencoders improves model performance in latent space
- > Derive **theoretical upper bounds** on W_2^2 and Cosine LogFC as functions of variance explain by PCA

Model Architecture

We train Flow Matching (FM) and Neural ODE models in the latent space of an autoencoder (AE) with standard MSE reconstruction loss and add geometric regularization loss, by matching diffusion (PHATE) potential distances in high dimensional gene-space with Euclidean distances in the autoencoder latent space

Both FM and Neural ODE models learn a velocity field which is integrated from control cells to generate predicted distributions of perturbed cells (see right box of diagram below).

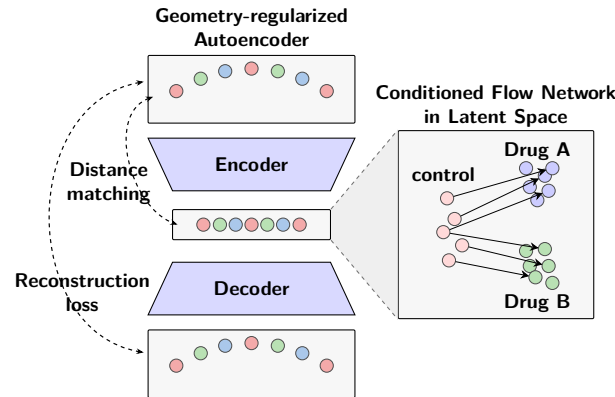


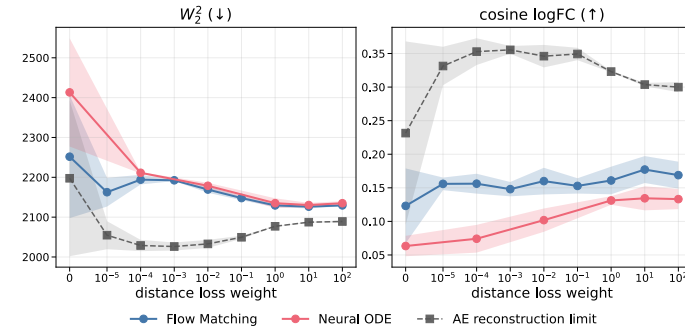
Diagram of geometrically regularized Autoencoder with distance and reconstruction loss. Showing flow model performed in latent space to predict perturbation (drug) effects.

Results

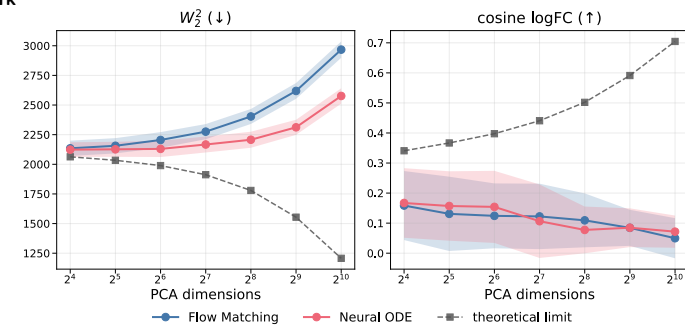
We find that adding geometric regularization to the autoencoder (AE) improve FM and ODE model performance across multiple metrics, when trained in the latent space of the AE. FM models consistently do better than ODE models on the Cosine LogFC metric and ODE models see a larger improvement from geometric regularization of the autoencoder.

We see that FM and ODE models do worse in PCA space as the dimensionality of the PCA space increases, even as the theoretical limit of the metric improves, which is counterintuitive.

FM + ODE performance vs AE geometric regularization strength



FM + ODE performance vs PCA dimensionality



Conclusion

We find geometric regularization improves FM + Neural ODE performance across metrics and stabilizes AE training, but the optimal regularization strength differs by metric.

We also find, counterintuitively, that flow model performance degrades in higher dimensional PCA space, even as the theoretical limit on model performance improves.

Takeaway 1: theoretical-limit analysis should be a standard design step for latent-space single-cell modeling

Takeaway 2: training space, loss, and evaluation metrics must be co-designed, not chosen independently

Future Work: multi-dataset benchmarking, dose-trajectory structure, prediction on completely unseen drugs

Model Evaluation Task

Underrepresented context generalization

Train Test

Biological context				
	Type 1	Type 2	Type 3	
$\otimes p_0$				(unperturbed)
$\text{⚡ } p_1$				Held-out underrepresented context
$\text{⚡ } p_2$				
$\text{⚡ } p_3$				
$\text{⚡ } p_4$				
$\text{⚡ } p_5$				